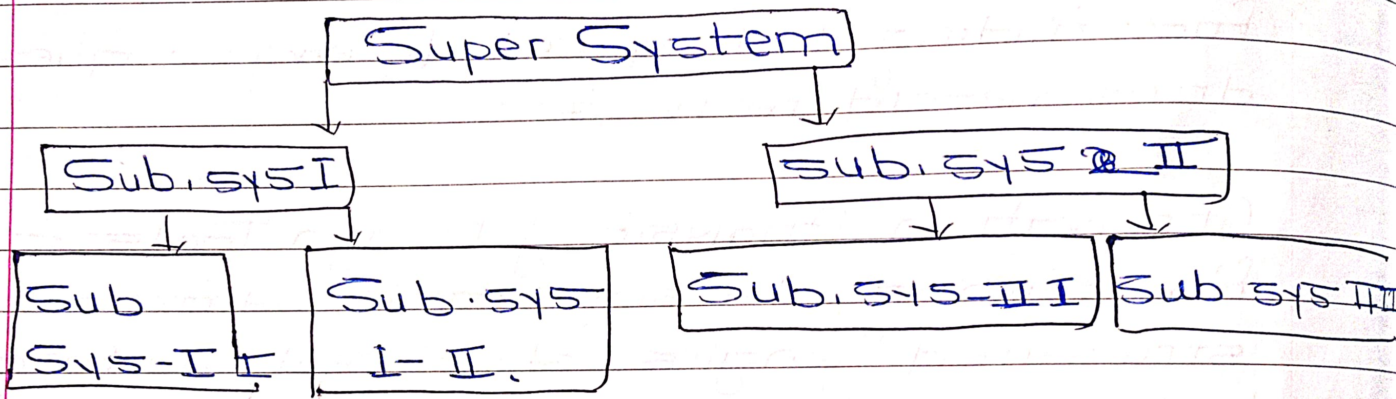
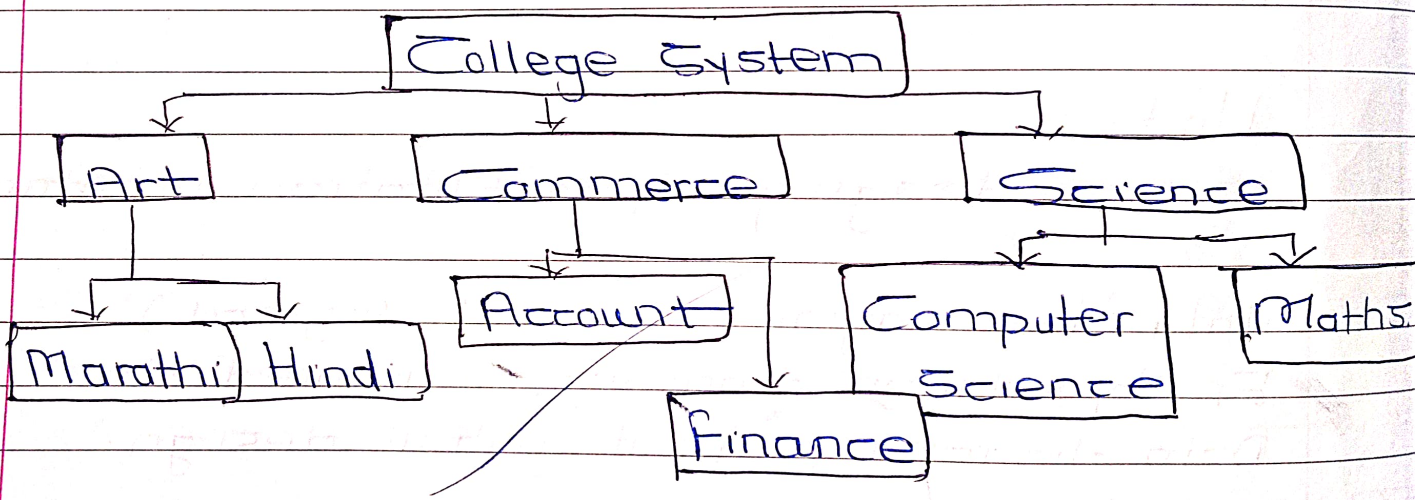
Introduction to Software Engineering

Defn 2M

System concept:- System is defined as it is a group of dependent component which are working together according to spefific goal or objective as called as system.





In above example, main system is college system & remaining are sub system like arts, commerce & science

In arts sub system there are two sub system like marathi & hindi

In commerce subsystem there are two sub system like account finance.

In science subsystem there are two sub system like computer science & maths.

IMP **Characteristics of system.**

1. the system is usable
2. Component of the system are interface to each other.
3. System has input & output
4. The system transfer input to output.
5. The system must be controlled.

IMP 5M **Elements of the system.**

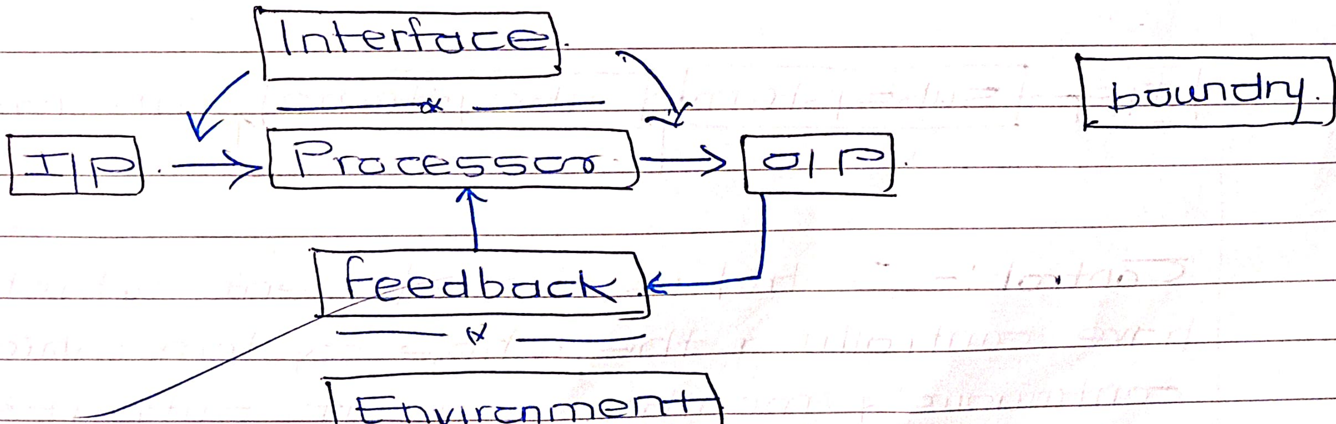


fig:- Elements of the system.

fig element of the system.

1. I/P
2. O/P
3. Processor / Process
4. Control
5. feedback
6. Boundry
7. Environment.

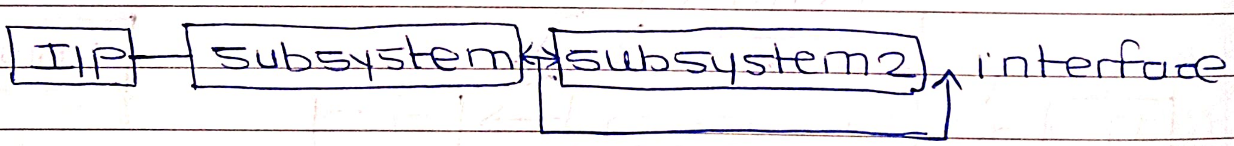
The elements of the system input output processor control feedback boundary & environment.

I/p:- These is the element which gets information from outside into the system for processing.

o/p:- These is the element which show the processed data or information from the input device.

Processor:- Processor which are used to process the data from input devices & transfer to output devices it is the interfacing between the two devices like input & output.

Interface:- Interface means communication link between the two devices.



Control:- Control is the element which have controlling the whole system which continuously monitering input, output & processor.

Feedback:- feedback is the element which is used to check the actual output with excepted output if there is some of the difference then it is interface to processing unit, the processor unit compared two input (input & output) & produces the excepted output

Boundary:- This is the element which indicate limit of the system temperature range 0°C to 100°C. It has 0° is the lower limit & 100°C is the upper limit.
If these limits are crosses the system goes in automatically shut down condition.

Environment:- Environment mean the elements outside the boundary or outside the system. is called as environment.

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Types of system:-

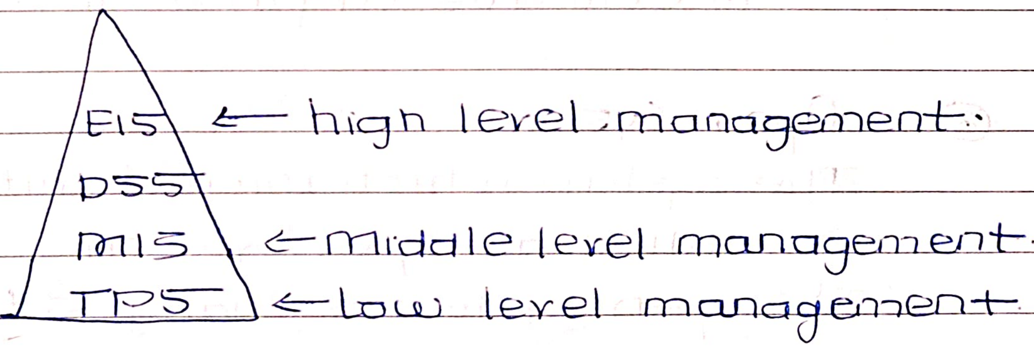

Fig: Type of system with management levels.

EIS - Execution Information system.
DSS - Decision support system.
MIS - Management information system.
TPS - Transition Processing system.

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① Deterministic System.

* This system operate predictable manner in this system output is known the output is fixed type the degree of error is not present a in the output. * the output generated by thes system is sure.

eg. The computer program to find out factorial of given number.

② Probablistic system.

This system operates in unpredictable manner in this system the output is not known in this system the output is not sure.

The degree of error is always present in the output. The fixed type of output is not come.

e.g. ① weather forecasting system.
② Coin tossed & it's outcome is not sure.

③ Open System.

The system which interact with the environment is called as open system.

The open system is changes their parameters with respect to environment if the environment temperature goes in decreasing state the effect show on human system & organization sys.

This system is totally depends on environment parameters.

④ Close System.

The system does not interact with environment is called as the closet system.

This system does not change their parameter

in response to environment.
e.g. chemical reactions.
A door sensor which automatically open & closed with the help of sensor.

⑤ TPS (Transition Processing System):-
(Transaction processing system)
Transaction means interaction takes place between the two element. the interaction between the bank account deposite or withdraw the money the TPS generated day by day reports and store the transitions of bank account.
It is primary use for generating transition reports, generating history of bank account & processing transition.
e.g. Banking system, cells accounting system.

Benefits of TPS.
① It stores all transitions
② It helps to find out the problems related to transitions
③ It generates current status reports of bank accounts.

⑥ MIS (management information system).
* It is use to support the manager in organization for planning & control the function.
The information store the systematically & also collect the new information
* The MIS generated the reports are meaningful.
* e.g. personal information system

Benefit's of MIS.
① It is suitable for analysis
② It is helpful for planning & control
③ It can handles equaries quickly.

⑦ DSS (Decision support system)
* The DSS is use for decision making with the support of low level system MIS.
It helps for decision making supporting with information collected by MIS
* It is use for summerising the data and collecting external data from the environment
* This system also use for analysis & planning modules such as management science & operation research
* This system mostly use by top level management for decision making.

Benefits of DSS
① Take better decision
② It saves court & time.

⑧ EIS (Execution information system).
It is structured & atomated tracking system
It provides accessing the information in management reports.
The system are user friendly because easily connected with online information service & email.

Benefits
① Easily & quick access to detail information
② Information provided in multiple views like text, graphical format, no. format.
③ It use filter & compressing the data.

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(4) It tracking the data in crehcal areq.

(9) Expert System.

* Expert system the human are expert in specific areas so their knowledge must be recorded for the future purple the expert system is helpful for use of human knowledge to solve the probler

* The component of expert system is knowledge aquire, knowledge base, user interface, explanitation faculity & knowledge defining of system.

Benefits of ES (Expert system)
(1) Output is selected with the opinion or many expert.
(2) It has use large knowledge base increase
(3) Output & productivity
(4) System is more flexible.

(10) System Analysis.

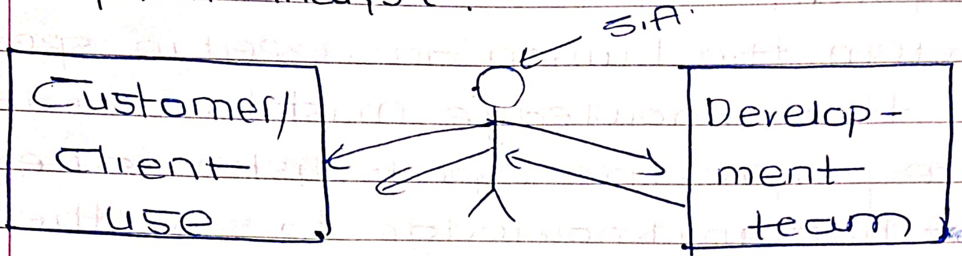
System analysis means identification, understandin & critically examining the system & it's sub sys for the purpose of achieving the output or result is called as system analysis.

The output is achieved through modificam chain in component of saperating the component upgresion in software & hardware.

System analysis performing different task:-
(1) Problem identification & understanding system
(2) Evolution & synthesis a system.
(3) Moduling & modification in the system
(4) Specification of the system.

⑤ Review of the system.

⑪ System Analyst:-



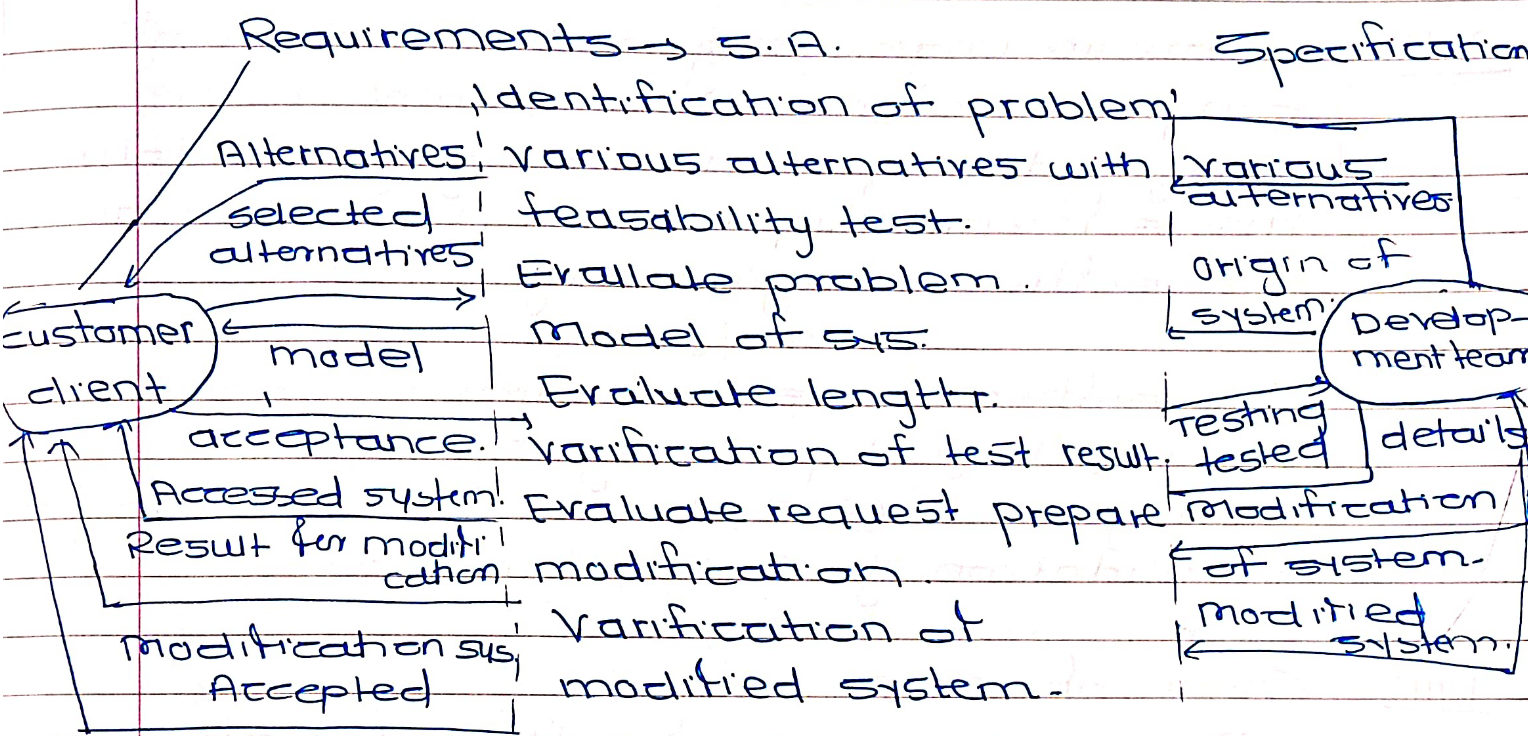
System analyst is the middle man betn the user or client or customer & software derelopment team is called as S.A.

Role of S.A.

The system analyst study system requirement through the use eyes & he has different roles like

1. The system analyst must be good communication skill to collect the information from user & represent the derelopment team.
2. The system analysis must be support the designing & programming team to execute different task.
3. System analyst must be find out & understand the exact problem with or without help of system developer.
4. System analyst must be understand requirement or customer & indentify the customer need.
5. The system analyst should be able to evolute the system.

Role paid by System Analyst. (S.A).



⑥ System analyst must be maintain good relationship with customer & development team

⑦ System analyst must be provide set of solutions to the problem.

⑧ System analyst find out the problems design the new system.

⑨ System analyst must be create protype or model of the understand data flow, control flow & processing of funn.

⑩ The system analyst must be capability of logical resing to produce new solution for the user problem.